

Decoding γ -ray signatures from core-collapse supernovae: First experimental constraints on the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate

M. Abubakar¹, J. S. Randhawa^{1,*}, S. R. Carmichael², P. D. O'Malley², D. W. Bardayan², J. J. Kolata²,
R. Longland³, C. Dembski², W. S. Porter², W. W. von Seeger², M. Sorensen², T. Psaltis⁴,
R. Zite² and M. A. Zubair¹

¹Department of Physics and Astronomy, *Mississippi State University*, Mississippi 39762, USA

²Department of Physics and Astronomy, *University of Notre Dame*, Notre Dame, Indiana 46556, USA

³*North Carolina State University*, Raleigh, North Carolina 27695, USA

⁴*Astronomy and Physics Department, Saint Mary's University*, Halifax, Nova Scotia B3H 3C3, Canada



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The γ -ray emitting radioisotopes produced during explosive Si-burning in core-collapse supernovae (CCSNe) could serve as a probe to supernovae explosion energetics. Especially important are the three isotopes ^{43}K , ^{47}Sc , and ^{59}Fe as these are potentially detectable with next generation space-based γ -ray telescopes. However, current nuclear physics uncertainties hinder model-observation comparisons. The $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction has been shown to be among the most important reactions which impact the production of these three γ emitters in CCSNe. We report on the first observation of resonant states in the compound nucleus ^{29}Si , which were populated via the $^{28}\text{Si}(d, p)^{29}\text{Si}$ reaction. This measurement provides the first experimental determination of the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate. This new measurement, along with associated uncertainty limits, removes one of the most important nuclear physics uncertainties in the production of ^{43}K , ^{47}Sc , and ^{59}Fe γ -emitting radioisotopes in CCSNe.

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Introduction. Core-collapse supernovae (CCSNe) are thought to be important sites for the synthesis of a wide range of elements from iron-peak to first-peak r -process elements, and for p nuclei [1,2]. The synthesis of elements and their injection into interstellar medium depends on the details of supernova explosion mechanisms [2,3]. Even after several decades of study, the exact details of the explosion mechanism and other physics of CCSNe still remain unknown. For example, the favored engine for supernova expansion invokes convection. As the explosion shockwave expands outward, most models predict that it quickly stalls. However, three-dimensional (3D) models have shown that the region between the proto-neutron star and the stalled shock is convectively unstable [4], which is able to revive the shock driving the explosion. There are various parameters in this engine which need to be constrained [4]. The explosion energy is one of the important parameters in the supernova engine as shown in previous studies [2–4]. Many radioisotopes produced in the CCSNe can provide useful insights into the underlying mechanism driving the explosion. A detailed study of Andrews *et al.* [3] shows that ^{43}K , ^{47}Sc , and ^{59}Fe are some of the most important γ -emitting radioisotopes that can directly probe the CCSNe explosion energy. For a given progenitor mass, the increase in shock energy increases the production of these radioisotopes, and hence this correlation can provide a direct probe of the explosion energy.

The next-generation γ -ray telescopes (e.g., COSI, e-astrogam [5,6]) can explore the soft γ -ray sky in the 0.2–5 MeV energy range. Thus, these radioisotopes can be studied by astronomers through MeV γ rays from the radioactive decay and nuclear de-excitations. Timmes *et al.* [7] estimated that with the future γ -ray telescopes, γ -ray signatures from many of these radioisotopes could be detectable out to a distance of 50 kpc and hence are promising candidates for future observations. Since production and survival of these isotopes depends on interplay of various nuclear reactions in CCSNe, it is important to understand and address the nuclear physics uncertainties to decode the observations. A recent study by Hermansen *et al.* [8] investigated the synthesis of 13 radioisotopes from ^{43}K to ^{59}Fe , and identified the key nuclear reaction rates to the nucleosynthesis of these radioisotopes. The sensitivity study of Hermansen *et al.* [8] has shown that $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ is one of the most important reactions that can directly impact the abundances of ^{43}K , ^{47}Sc , ^{59}Fe .

These isotopes are produced in explosive silicon burning, where the relevant temperature range for the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction is from 2–2.4 GK [8], and the Gamow window is from 0.5 MeV to 1.8 MeV [8]. The $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction proceeds through resonant states in the $p + ^{28}\text{Al}$ system, i.e., the ^{29}Si compound nucleus. The Gamow window for this reaction corresponds to an excitation energy range of interest in ^{29}Si from 12.8–14.1 MeV, respectively. Currently, there are no experimental constraints on this reaction rate. In compound nuclei where the level densities are sufficiently high and/or resonances become broader, levels overlap considerably such

*Contact author: jsr512@msstate.edu

that there is little structure left in the cross section. In such cases, statistical models are applied where cross sections are described by averaged quantities. Currently $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction cross section estimates are based on statistical-model based Hauser-Feshbach (HF) calculations (one of the most commonly applied methods). For (p, α) reactions, the statistical model calculations mainly require level density (LD) and optical model potential (OMP) parameters. Therefore, any uncertainty in these inputs lead to uncertainty in the calculated cross sections. Moreover, it is difficult to ascertain in some cases whether the statistical model is applicable or not. For the current case of ^{29}Si at higher excitation energies, high level-densities are expected, and statistical models may be appropriate. In neighboring nuclei, however, isolated resonances have still been shown to dominate the reaction rate and significant deviations from the statistical model predictions exist. For example, the $^{22}\text{Mg}(\alpha, p)^{25}\text{Al}$ reaction rate derived from both direct and indirect measurements (i.e., exploring resonant states in ^{26}Si) are lower than the statistical model predictions [9,10].

Even though a direct measurement of the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ in the upper end of Gamow window might be possible at radioactive ion beam facilities, the predicted cross sections (from statistical models) in the lower half of Gamow window are too low for a direct measurement with the projected beam intensities at current facilities ($\leq 10 \mu\text{b}$ below 1 MeV center-of-mass energies). Therefore, indirect measurements must be implemented to provide valuable constraints on this reaction rate in the Gamow window. In this Letter, we report the first experimentally determined reaction rate for the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction from a high-resolution spectroscopy study of ^{29}Si through the $^{28}\text{Si}(d, p)^{29}\text{Si}$ reaction. Details of the experiment, the derived reaction rate, and the impact on abundances of three radioisotopes of interest are shown in the subsequent sections.

Experiment details and data analysis. A high-resolution spectroscopy study of ^{29}Si was performed by populating the nucleus of interest through the $^{28}\text{Si}(d, p)^{29}\text{Si}$ transfer reaction, i.e., a neutron pick-up reaction. The experiment was performed using the newly commissioned Enge Split-Pole Spectrograph in the Nuclear Science Laboratory (NSL) at the University of Notre Dame [11,12]. The reaction Q value is 6.249 MeV, and the beam energy was chosen to populate the states of interest in ^{29}Si from 12.8–14.2 MeV (i.e., in the Gamow window for CCSNe). A deuteron beam of 14 MeV was impinged on a $\approx 700 \mu\text{g}/\text{cm}^2$ ^{28}Si target. Data were collected at 11 different entrance angles of the spectrograph ranging from 12° – 32° in the laboratory frame. Reaction products (charged particles) entering the spectrograph were separated and focused onto the focal plane based on their magnetic rigidity.

The focal plane detector was a position sensitive ionization chamber, which consisted of a cathode and two position-sensitive anode sections. Each position sensitive section consists of 220 conducting pads on a printed circuit board (PCB) and three anode wires that run parallel to the PCB. Each conduction pad has a width of 2.286 mm and is separated by 0.254 mm [13]. Ionization electrons are picked by anode wires, inducing charge on pick-up pads. Each pick-up pad

is attached to a delay tap that delays the signal by 5 ns and each delay tap is connected in series to form a delay line. The induced charge on the pick-up pads travels to two ends of the delay line. The timing difference between the signals on either end of the delay line provided a position measurement of the particle. In addition, a plastic scintillator was placed behind the focal-plane detector, to measure their residual energy. At a deuteron bombarding energy of 14 MeV, the primary reaction channels open are elastic/inelastic scattering, $^{28}\text{Si}(d, p)^{29}\text{Si}$ and $^{28}\text{Si}(d, \alpha)^{26}\text{Al}$. Therefore, to identify reaction products of interest, particle identification was obtained using the energy deposited in the cathode versus the focal-plane position, and alternatively from the energy deposited in the cathode versus the residual energy in the scintillator, where either of these two methods worked equally well. By selecting protons, the ^{29}Si excitation energy spectrum was obtained. To determine the measured excitation energies of the observed proton peaks, the positions of events as measured by the focal plane detector provided the radius of curvature of events. Using the magnetic field values as obtained from a nuclear magnetic resonance (NMR) probe, the $B\rho$ values were extracted which were directly correlated with the momenta (and hence energies) of the protons. Alternatively, since the bound excited states in ^{29}Si are very well studied, populated known levels in the residual nuclei were used for calibration. Because the momenta of the particles were known, a direct calibration between the front time difference (position) and the ejectile momentum was performed. Excitation energies from these two complimentary methods agree within 4 keV, which was used as a part of final uncertainty in the resonance energies. Since the bound excited states in ^{29}Si are known, peaks seen in excitation spectrum were all attributed to known states and no spurious states were seen, thus eliminating the possibility of any target contamination. In the excitation energy spectrum, an energy resolution of the order of ≈ 30 keV was obtained. To extract the angular distribution, protons were measured at laboratory angles from 12° to 32° .

Results. The ^{29}Si excitation energy spectra are shown in Fig. 1. The top panel shows excitation energies up to 8.5 MeV at two different angles, a region where states have been well studied in various studies [14], whereas, the bottom panel shows the proton-unbound states observed in this study. The inset in the top panel shows the measured angular distribution for the ground state.

To extract spectroscopic factors, angular distributions were compared with distorted wave Born approximation (DWBA) calculations using the code FRESKO [15]. Optical model parameters and neutron bound-state form factors were chosen following the approach of Peterson *et al.* [14] and adjusted for the present beam energy. The optical-model parameters used by Peterson *et al.* for the deuteron and proton potentials were taken from Perey and Perey [16] and Becchetti and Greenlees [17], respectively. These are both energy-dependent global potentials which provide excellent fits to the corresponding elastic scattering data throughout the energy ranges of interest. For population of the ground state, the extracted spectroscopic factor was 0.49 ± 0.04 , which agrees with previously reported values of 0.51 and 0.53 reported in Ref. [14]. The quality of the fit to the present ground-state data (Fig. 1, upper panel), for

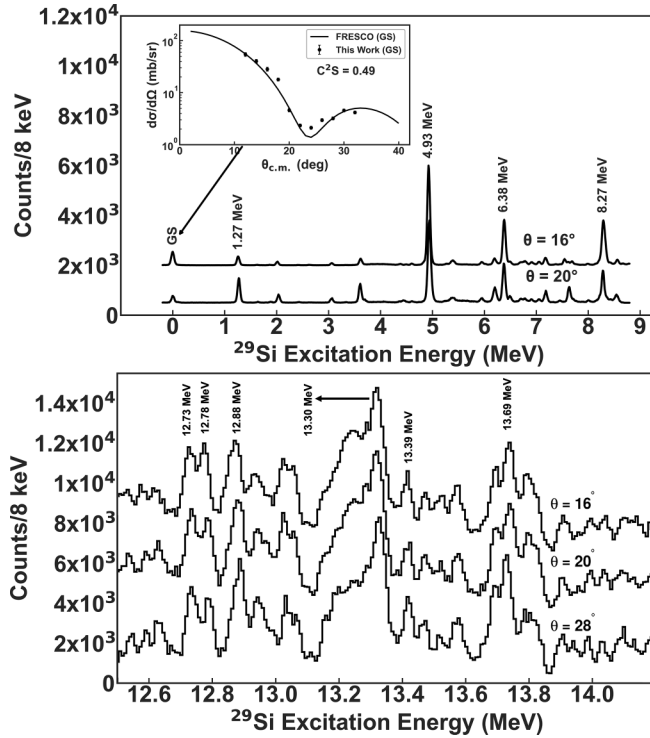


FIG. 1. Top panel: Spectra of (well-studied) bound states in ^{29}Si . Bottom panel: Spectra showing proton-unbound ($S_p = 12.33$ MeV) states in ^{29}Si where most of the states were observed for the first time, and some of the prominent states are labeled.

both the angular distribution and spectroscopic factor testifies to the applicability of these parameters to the $^{28}\text{Si}(d, p)^{29}\text{Si}$ reaction at the current energy. Bound neutron states were described by Woods-Saxon potentials with physical separation energies. Unbound states were treated using discretized continuum neutron overlaps with negative separation energies.

The bottom panel of Fig. 1 shows the ^{29}Si excitation energy spectrum from 12.4 MeV–14.2 MeV at three different angles. States with energies between 12.8–14.1 MeV could provide resonances in the Gamow window for the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction. There is very little data at these higher excitation energies. Peterson *et al.* [14] observed four excited states in the relevant Gamow window [i.e., E_x (in MeV)=13.05, 13.44, 13.75, 13.94]. However, it is difficult to critically analyze any of these states when no detailed spectra at higher excitation energies were provided.

In this study, a total 31 resonances were identified with 26 lying in the Gamow window. At these excitation energies, no sharp $\ell = 0$ or 1 transitions are expected as also pointed out and shown in Peterson *et al.* [14]. No broad levels were observed in the present study as well. Angular distributions were extracted for all the resonant states and possible ℓ values were extracted from comparisons with Fresco calculations for $\ell = 2, 3$, and 4. The extracted angular distributions and comparison to the calculations are shown in Fig. 2, where only a few example states are shown for clarity. There was some ambiguity concerning the angular distributions of some states best fit with an $\ell = 2$ transfer because unreasonable

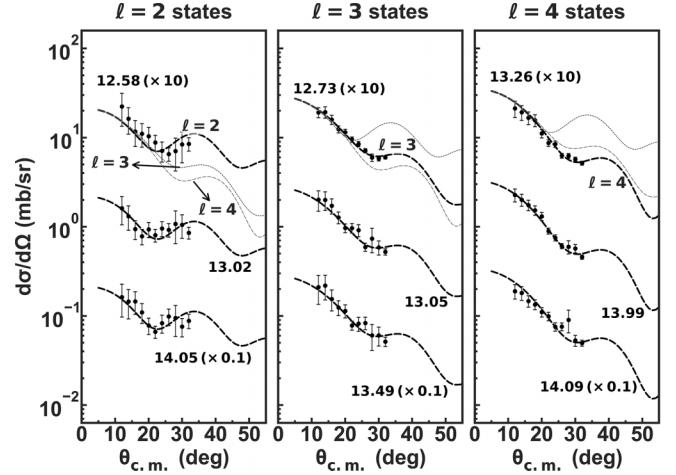


FIG. 2. The angular distributions for populating representative states scaled arbitrarily for display purpose. Fresco calculations for different angular momentum transfers are also plotted for comparison.

spectroscopic factors were required to fit the data, and $\ell = 3$ angular distributions could not be excluded. Therefore, some of these $\ell = 2$ states could alternatively be $\ell = 3$ transfers. For any state if there is an ambiguity in the angular distributions, the assigned ℓ values are tentative and are shown in braces in Table I. The ℓ values were then used to calculate the penetrabilities to get the proton and α -particle partial widths. These considerations lead to a factor of ≈ 1.6 uncertainty in the reaction rate and are accounted for in the final uncertainty.

To account for uncertainties in the calculated quantities and to propagate these uncertainties to the calculated reaction rate, we used the code RATESMC [18], which has been extensively used to calculate reaction rates and associated uncertainties using a Monte Carlo method. To use this code and properly account for associated uncertainties, we have assumed total 7 keV uncertainty in the resonance energies which reflects quadrature sum of two contributions, 6 keV from the maximum change in the resonance energies across whole angular range covered in this study and 4 keV from focal plane detector calibration as discussed earlier. To calculate resonance strengths (and associated uncertainties), we have followed the approach of Longland *et al.* [18], where when upper limits of partial widths are provided, the code samples over a Porter-Thomas distribution of dimensionless reduced widths with a mean value of $\langle \theta^2 \rangle$. We calculated the upper limit of partial widths using $\theta^2 = 0.3$ for protons using Iliadis *et al.* [19] and 1.0 for α s, with $C^2S = 1$ for both protons and α particles. The mean values for dimensionless reduced widths $\langle \theta^2 \rangle$ for protons and α s used were 0.0045 and 0.03, respectively, following Ref. [20]. Table I shows the excited states used in reaction rate calculations together with calculated proton and α partial widths, which were treated as upper limits in the code RATESMC.

Impact on γ -ray emitting isotopes in CCSNe. The calculated reaction rate is shown in Fig. 3 (upper panel) (blue dotted line) along with the uncertainty band (blue band). In comparison, we have shown the rate from REACLIB [21]

TABLE I. Selected resonance parameters: excitation energy E_x , resonance energy E_r , orbital angular momentum ℓ transferred in the (d, p) reaction, and partial widths Γ_p and Γ_α . Values given in parenthesis show tentative assignments.

E_x (MeV)	E_r (keV)	ℓ transfer	Γ_p (eV)	Γ_α (eV)
12.58	250	2	7.8×10^{-4}	2.3×10^{-1}
12.63	300	3	1.5×10^{-4}	8.2×10^{-2}
12.67	340	(3)	6.7×10^{-4}	1.3×10^{-1}
12.73	400	3	4.1×10^{-3}	2.5×10^{-1}
12.78	450	3	1.4×10^{-2}	4.1×10^{-1}
12.88	550	3	1.0×10^{-1}	1.1
12.93	600	2	7.6	7.9
12.97	640	(4)	6.5×10^{-3}	3.4×10^{-1}
13.02	690	2	2.4×10^1	1.7×10^1
13.05	720	3	1.1	4.6
13.10	770	(2)	5.7×10^1	3.1×10^1
13.16	830	(2)	9.9×10^1	4.7×10^1
13.23	900	(3)	6.9	1.7×10^1
13.26	930	(4)	1.9×10^{-1}	3.3
13.30	970	3	1.2×10^1	2.8×10^1
13.39	1060	2	5.4×10^2	2.1×10^2
13.44	1110	2	7.3×10^2	2.8×10^2
13.49	1160	3	4.6×10^1	8.9×10^1
13.54	1210	(2)	1.3×10^3	4.8×10^2
13.60	1270	3	8.6×10^1	1.6×10^2
13.66	1330	3	1.2×10^2	2.2×10^2
13.69	1360	4	4.1	4.4×10^1
13.75	1420	3	1.8×10^2	3.4×10^2
13.77	1440	(3)	2.0×10^2	3.8×10^2
13.86	1530	4	1.0×10^1	1.0×10^2
13.90	1570	3	3.5×10^2	6.8×10^2
13.94	1610	3	4.1×10^2	8.1×10^2
13.99	1660	4	1.8×10^1	1.9×10^2
14.05	1720	2	9.2×10^3	4.6×10^3
14.09	1760	4	2.8×10^1	2.9×10^2
14.10	1770	(2)	1.1×10^4	5.6×10^3

which uses statistical model based Hauser-Feshbach calculations from the NON-SMOKER code (red dashed line), and the uncertainty assumed in that rate from Ref. [8] (i.e., a factor of 100 up and down). The new reaction rate has roughly a factor of ≈ 3 uncertainty (up and down), much less compared to the previous uncertainty of a factor of 100 up and down, hence representing a large improvement compared to previous uncertainties. The lower panel in Fig. 3 shows the ratio of NON-SMOKER predictions to the current rate. The current rate agrees with the NON-SMOKER (REACLIB) predictions in the lower end of Gamow window within the uncertainty limits.

To understand the impact of the new reaction rate and the associated uncertainty on the γ -emitting radioisotopes in CCSNe, we used information from the Hermansen *et al.* [8] sensitivity study. Figure 4 illustrates the astrophysical impact of the new $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate measurement on the abundances of the three γ -emitting radioisotopes of interest (^{43}K , ^{47}Sc , and ^{59}Fe) in CCSNe. In Hermansen *et al.* [8] they used the postprocessing nuclear reaction network (SkyNet) to model the explosive burning in the CCSNe for a $12M_\odot$

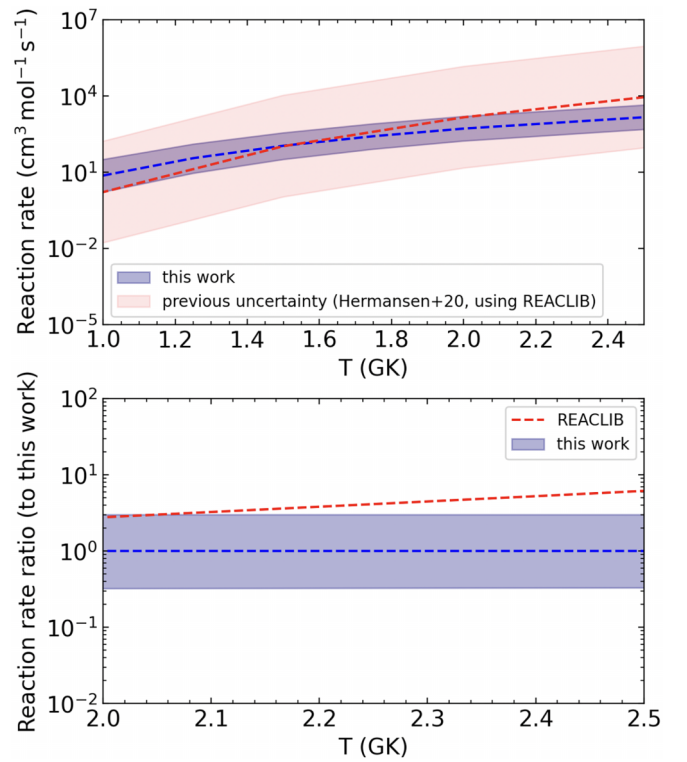


FIG. 3. Reaction rate derived in this work. Top panel: Comparison of reaction rate from current work to that from REACLIB (NON-SMOKER) used in Hermansen *et al.* [8]. Lower panel: Ratio to current reaction rate, zoomed into Gamow window i.e. from 2 GK–2.4 GK. Lower panel shows that current reaction rate agrees with REACLIB (NON-SMOKER) rate, in the lower end of Gamow window, within the uncertainty limits.

progenitor. The REACLIB reaction rate library was used where the default $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate was based on HF NON-SMOKER calculations. This $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate was systematically varied using multiplicative factors of 0.01, 0.1, 0.5, 2, and 100, and the resulting change in isotopic abundances was noted. We plotted the abundance ratios, which is the abundance of each isotope relative to the baseline value (i.e., when the multiplication factor is 1) as a function of the reaction rate multiplier to show the sensitivity of ^{43}K , ^{47}Sc , and ^{59}Fe abundances to the variation in REACLIB rate. Figure 4 shows that with a factor of 100 change in this reaction rate, ^{43}K and ^{47}Sc abundances can change by factors of 4 and 2, respectively. ^{43}K shows the highest sensitivity to changes in the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate followed by ^{47}Sc and ^{59}Fe . However, if the rate is lower than the NON-SMOKER rate (e.g., multipliers of 0.5, 0.1, 0.01), there is little to no change in the abundance. As shown in Fig. 3, our rate (within the current uncertainty limits) agrees with default rate in REACLIB at lower temperatures and is lower than REACLIB at higher temperatures. The current measurement thus rules out any enhancement with respect to default REACLIB rate and hence rules out multiplication factors 2, 10, and 100, where the abundance shows the maximum sensitivity. The new uncertainty limits on this reaction rate are shown as the width of the vertical band in Fig. 4. With the new $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction rate

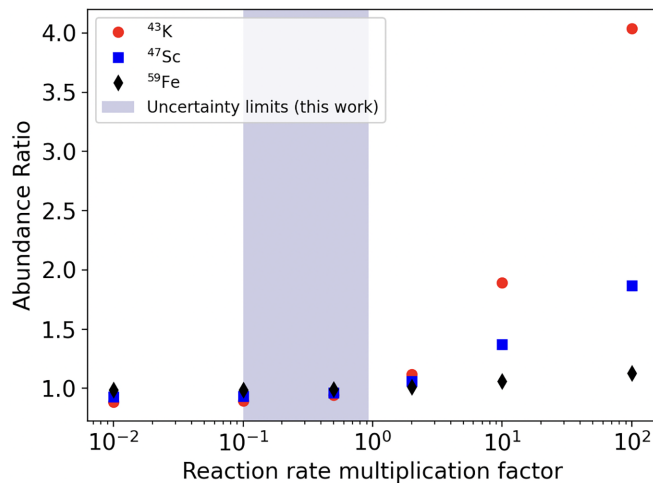


FIG. 4. Impact of current measurement on the abundances of γ -ray emitters in CCSNe. y axis shows the abundance ratio for ^{43}K , ^{47}Sc , and ^{59}Fe for $^{28}\text{Al}(p, \alpha)^{25}\text{Mg} \times \text{multiplier}$ to baseline rate from REACLIB, where rate multiplier is shown on x axis.

and current uncertainty limits, there is little to no change in the abundances of ^{43}K , ^{47}Sc , and ^{59}Fe in the CCSNe. The current measurement thus removes an important nuclear physics uncertainty in the production of three major γ -emitting radioisotopes in the CCSNe and hence increases the fidelity of future models to predict the future γ -ray observations from CCSNe.

Summary. To summarize, next-generation γ -ray telescopes like COSI, planned to be launched in 2027, can explore the soft γ -ray sky in the 0.2–5 MeV energy range with higher sensitivities compared to COMPTEL or INTEGRAL/SPI [5]. Yields of radioisotopes from astrophysical models of CCSNe show that with next-generation telescopes, γ rays from nuclear de-excitations of isotopes produced can be observed out to distances of up to 50 kpc [7]. Detecting isotopes such as ^{43}K and ^{47}Sc produced in explosive Si-burning in CCSNe can directly probe the explosion energy and mixing in the CCSNe [3].

Sensitivity studies have shown that the production of these γ -emitting isotopes is strongly affected by uncertainties in several reaction rates [8]. Among some of the most influential of these is the $^{28}\text{Al}(p, \alpha)^{25}\text{Mg}$ reaction, which impacts the abundances of ^{43}K , ^{47}Sc , and ^{59}Fe . Prior to the present work, very little experimental information existed. The reaction rate

sensitivity study by Hermansen *et al.* [8] assumed an uncertainty of up to a factor of 100. We have provided the first experimental constraints on resonant states in the $p + ^{28}\text{Al}$ system, i.e., the ^{29}Si compound nucleus. The states of interest were populated via the $^{28}\text{Si}(d, p)^{29}\text{Si}$ transfer reaction at the University of Notre Dame, using a deuteron beam and analyzing the reaction products with the Enge split-pole spectrograph. We identified 26 new resonances in the Gamow window. The reaction rate derived from this experimental information agrees with statistical-model predictions within uncertainties at the lower end of the Gamow window (using the upper uncertainty bound), and is up to a factor of 2 lower at the higher end of the window. Within these new uncertainty limits the updated reaction rate does not impact the production of ^{43}K , ^{47}Sc , or ^{59}Fe in CCSNe. These measurements therefore remove an important nuclear-physics uncertainty in CCSNe models to predict the future γ rays from CCSNe.

Outlook. With next-generation γ -ray telescopes being scheduled for launch in coming years and expected to provide improved sensitivity to MeV range γ rays, prospects of detecting γ rays from a future galactic CCSNe are exciting. With new platforms like the Supernova Early Warning System (SNEWS) [22], sensitive neutrino detectors like Super-Kamiokande [23] can alert the electromagnetic counterparts for early observations, as the arrival of burst of neutrinos will be the first indication of a potential CCSNe. These early warnings and observations in early phases will be important to decipher any γ rays from ^{43}K decay as it has a short half-life (22.28 h). Any signature or absence of ^{43}K signature can provide stringent constraints on the astrophysical models of CCSNe, provided the nuclear physics uncertainties are addressed. Therefore, the current uncertainties in other reaction rates, to which ^{43}K shows high sensitivity, also need to be addressed. Some of these high-impact reactions for ^{43}K , as shown in Hermansen *et al.* [8] include $^{42}\text{K}(n, \gamma)^{43}\text{K}$, $^{43}\text{K}(p, n)^{43}\text{Ca}$, and $^{42}\text{K}(p, n)^{42}\text{Ca}$, which involve reactions on radioisotopes, well within the reach of advanced radioactive ion beam facilities.

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Data availability. The data are available from the authors upon reasonable request.

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